G NIHAL

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Software Developer and Computer Science and Engineering student focused on building practical web applications and full-stack software solutions. Hands-on experience developing full-stack projects across frontend interfaces, backend APIs, databases, and authentication. Continuously strengthening software engineering and problem-solving skills through Data Structures and Algorithms and real-world project development.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Python, Java, C#, SQL, HTML5, CSS
Frontend	React.js, Redux, React Native, Tailwind CSS, Responsive Web Design
Backend	Node.js, Express.js, .NET (Basics), REST APIs, JWT Authentication
Databases	PostgreSQL, MongoDB, MySQL
CS Fundamentals	Data Structures, Algorithms, OOP, DBMS
Tools	Git, GitHub

PROJECTS

SabAI Pay – AI-Powered UPI Payment Assistant

Tech Stack: React, Node.js, Express, Supabase PostgreSQL, Google Gemini AI, Razorpay, JWT

- Built a prototype AI-powered UPI payment assistant demonstrating Agent Pay and Reserve Pay through conversational, policy-constrained agentic payments
- Designed a separation-of-concerns architecture where Gemini proposes structured intents, a deterministic policy engine authorizes payments, and the payment engine executes auditable transactions
- Implemented AI chat flows for UPI payments, mobile recharge, bill payments, food ordering, auto-pay scheduling, and SabAI Gems cashback with Reserve Pay spending limits
- Integrated JWT authentication, mock bank/UPI flows, Razorpay test-mode payments, transaction auditing, and multi-payment method support across UPI, bank account, Reserve Pay, and SabAI Gems

Repository: github.com/nihal705/sabai-pay

MyCodeNotes – Full Stack Learning & Coding Platform

Tech Stack: React, Tailwind CSS, Python, FastAPI, PostgreSQL, SQLAlchemy, JWT, pgvector, RAG, Groq, Gemini

- Built a structured learning platform with four searchable, filterable modules — 80+ solved LeetCode problems, 250+ practice problems, 200+ concept references, and 9+ personal notes — with dual-language (Java/Python) solutions across Easy/Medium/Hard tiers
- Engineered NextraAI, a domain-specific AI assistant using retrieval-augmented generation (RAG) over 2,500+ pgvector-embedded content chunks, with tiered knowledge routing, SSE streaming, source citations, per-user rate limiting, and Groq → Gemini model fallback
- Developed a FastAPI + PostgreSQL REST backend (~50 endpoints, 17 SQLAlchemy-modeled tables) with JWT authentication, bcrypt password hashing, Pydantic validation, and email OTP signup — powering progress tracking, streaks, AI study plans, and timed mock tests
- Designed a responsive React frontend with dark/light mode, Framer Motion animations, a full-screen chat interface with voice input and multi-source context pinning, and PDF/CSV progress export, deployed on Vercel and Render with Git-based automatic deployments

Live Demo: mycodenotes.vercel.app | Repository: github.com/nihal705/MyCodeNotes

Weather App – Responsive Weather Forecast Application

Tech Stack: React, JavaScript, CSS, OpenWeather API, Vite

- Built a responsive weather application with city search, current conditions, and a 5-day forecast
- Integrated the OpenWeather API to surface six live metrics per city: temperature, feels-like temperature, condition, humidity, wind speed, and atmospheric pressure
- Implemented loading and error-state handling with reusable React components and a glassmorphism-style responsive UI

Live Demo: weather-app-nxhl.vercel.app | Repository: github.com/nihal705/weather-app

EDUCATION & CERTIFICATIONS

Bachelor of Engineering (B.E.) – Computer Science & Engineering *2023 – 2027*

S.E.A College of Engineering & Technology

Coursework: DSA, DBMS, OS, CN, Software Engineering

Certifications: React and Redux | From Relational Model (SQL) to MongoDB's Document Model | Ethical Generative AI in Software Engineering

ADDITIONAL INFORMATION

Languages: English, Kannada, Hindi